

ASSIGNMENT #05

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
Term: Fall 2025
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Due Date: Thursday, October 30th my Office (06:00PM)

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

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CHE 301

Assignment Number 2

October 30, 2025

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1. **Problem A:** The following *virial* equation of state (originally developed by Beattie and Bridgeman, *Proc. Am. Acad. Arts Sci.*, **63**, 229–308 (1928)) was proposed by Deming and Shupe, *J. Am. Chem. Soc.*, **52**, 1382–1389 (1930), for the prediction of the thermal properties of Nitrogen in the temperature range $[-70^{\circ}\text{C}, 400^{\circ}\text{C}]$ and in the pressure range $[0 \text{ atm}, 400 \text{ atm}]$:

$$PV^2 = RT \left[V + B_0 \left(1 - \frac{b}{V} \right) \right] \left(1 - \frac{c}{VT^3} \right) - A_0 \left(1 - \frac{a}{V} \right) \quad (1)$$

where $\{a, A_0, b, B_0, c\}$ are numerical constants that were determined from experimental data.

The values reported by Deming and Shupe are the following when P is the pressure, in atm, V the specific volume, in g/cm^3 , and R the universal gas constant, in $\text{cm}^3 \cdot \text{atm}/(\text{g} \cdot \text{K})$.

Table 1: Virial Equation Constants

$A_0 = 1528.6$	$a = 0.2748$	$c = 2.2 \times 10^6$
$B_0 = 1.643$	$b = -0.9235$	

Later in 1952, Lunbeck, Michels and Wolkers, *Appl. Sci. Res.*, **A3**: 197 (1952), reported Z the compressibility factor of Nitrogen at very high pressures. The data are shown in the table below:

Table 2: Compressibility Factor Z of Nitrogen

P (atm)	Temperature ($^{\circ}\text{C}$)				
	0	25	50	75	100
0	1.000	1.000	1.000	1.000	1.000
1	1.000	1.000	1.000	1.000	1.000
3	0.999	0.999	1.000	1.000	1.001
5	0.998	0.998	0.999	1.000	1.001
10	0.996	0.998	1.001	1.001	1.002
30	0.994	0.996	1.004	1.007	1.009
50	0.993	0.996	1.004	1.007	1.009
100	0.984	1.004	1.018	1.027	1.033
200	1.034	1.074	1.102	1.124	1.143
300	1.134	1.146	1.154	1.160	1.164
400	1.256	1.254	1.253	1.250	1.247
600	1.524	1.495	1.471	1.450	1.432
800	1.798	1.723	1.697	1.658	1.625
1000	2.067	1.988	1.923	1.867	1.819

The following modification of the Deming–Shupe *virial* equation is proposed to include the new high-pressure data:

$$PV^2 = RT \left[V + B_0 \left(1 - \frac{b}{V} \right) \right] \left(1 - \frac{c}{VT^3} \right) - A_0 \left[1 - \frac{a}{V} - \frac{d}{V^2} \right] \quad (2)$$

where d is a new numerical constant to be determined by the high-pressure data.

Calculations to be performed:

1. **Develop a methodology** to estimate the new constant d from the high-pressure experimental data by minimizing the following objective function:

$$\Phi(d) = \min \sum_i (P_i^{\text{exp}} - P_i^{\text{model}})^2 \quad (3)$$

Hint: Recast the virial equation of state in the form

$$Z = 1 + \beta\rho + \gamma\rho^2 + \delta\rho^3 \quad (4)$$

where Z is the compressibility factor, ρ is the molar (or mass) density of nitrogen, and $\{\beta, \gamma, \delta\}$ are coefficients that depend on temperature and on the constants $\{a, A_0, b, B_0, c, d\}$.

For the estimation of the new constant d , keep the values of the rest of the constants $\{a, A_0, b, B_0, c\}$ the same as given in Table 1 above. Prepare two **parity plots**, the first with $d = 0$ and the second with the optimal value calculated by nonlinear regression.

2. Calculate the thermal properties of Nitrogen $\{\beta, C_p, C_v, \mu_{JT}, v_{\text{sound}}\}$ at temperature $T = -60^\circ\text{C}$ and pressure $P = 800\text{ atm}$ using the modified virial equation. Express your answers in SI units. Compare your results with those obtained using the **(RKS)** Redlich–Kwong–Soave Equation of State and the values reported in the NIST Webbook.
3. Calculate the power required to compress Nitrogen reversibly and adiabatically from $T = -60^\circ\text{C}$ and $P = 800\text{ atm}$ to $P = 1000\text{ atm}$. Express your result in kJ/kg and compare it with the one obtained using the **(RKS)** Redlich–Kwong–Soave Equation of State.
4. Verify whether the Beattie–Bridgeman Equation of State is consistent at the critical point of nitrogen. If based on the deviations from consistency it is not, make a recommendation on how to rectify the inconsistency.

Ans:

1. In this question, we were tasked with optimizing the constant d for the Deming-Shupe virial equation. The first step in this process is to take the given virial equation and convert it to an equation in terms of compressibility, Z . To do this, we multiply both sides of the Deming-Shupe virial equation by a factor of $\frac{1}{VRT}$:

$$\left[PV^2 = RT \left[V + B_0 \left(1 - \frac{b}{V} \right) \right] \left(1 - \frac{c}{VT^3} \right) - A_0 \left(1 - \frac{a}{V} - \frac{d}{V^2} \right) \right] \frac{1}{VRT}$$

After distributing and canceling out like terms, we get:

$$\frac{PV}{RT} = \frac{1}{V} \left[V + B_0 - \frac{B_0 b}{V} \right] \left[1 - \frac{c}{VT^3} \right] - \frac{A_0}{VRT} \left(1 - \frac{a}{V} - \frac{d}{V^2} \right)$$

Since the $Z = \frac{PV}{RT}$, we can now rewrite the equation in terms of Z:

$$Z = \frac{1}{V} \left[V + B_0 - \frac{B_0 b}{V} \right] \left[1 - \frac{c}{VT^3} \right] - \frac{A_0}{VRT} \left(1 - \frac{a}{V} - \frac{d}{V^2} \right)$$

This equation can further be written as:

$$Z = 1 - \frac{c}{VT^3} + \frac{B_0}{V} - \frac{B_0 c}{V^2 T^3} - \frac{B_0 b}{V^2} + \frac{c B_0 b}{V^3 T^3} - \frac{1}{RT} \left(\frac{A_0}{V} - \frac{A_0 a}{V^2} - \frac{A_0 d}{V^3} \right)$$

Since $\rho = \frac{1}{V}$, we can write the equation in terms of ρ :

$$Z = 1 - \frac{c}{T^3} \rho + B_0 \rho - \frac{B_0 c}{T^3} \rho^2 - B_0 b \rho^2 + \frac{c B_0 b}{T^3} \rho^3 - \frac{1}{RT} (A_0 \rho - A_0 a \rho^2 - A_0 d \rho^3)$$

After combining like terms, we can then write the equation as:

$$Z = 1 + \rho \left(-\frac{c}{T^3} + B_0 - \frac{A_0}{RT} \right) + \rho^2 \left(-\frac{B_0 c}{T^3} - B_0 b + \frac{A_0 a}{RT} \right) + \rho^3 \left(\frac{c B_0 b}{T^3} + \frac{A_0 d}{RT} \right)$$

To simplify the equations for further calculations, we wrote the equation as:

$$Z = 1 + \beta(T)\rho + \gamma(T)\rho^2 + \delta(T)\rho^3$$

where:

$$\beta(T) = \left(-\frac{c}{T^3} + B_0 - \frac{A_0}{RT} \right)$$

$$\gamma(T) = \left(-\frac{B_0 c}{T^3} - B_0 b + \frac{A_0 a}{RT} \right)$$

$$\delta(T) = \left(\frac{c B_0 b}{T^3} + \frac{A_0 d}{RT} \right)$$

To do the minimization of pressure, we rewrote the equation for Z in terms of Pressure, P. We use the equation: $Z = \frac{P}{\rho RT}$ to rewrite the function of Z in terms of P:

$$Z = 1 + \beta(T)\rho + \gamma(T)\rho^2 + \delta(T)\rho^3$$

$$Z = \frac{P}{\rho RT} = 1 + \beta(T)\rho + \gamma(T)\rho^2 + \delta(T)\rho^3$$

$$P = RT [\rho + \beta(T)\rho^2 + \gamma(T)\rho^3 + \delta(T)\rho^4]$$

Using the above derivations, we were able to compile a function in VBA. This function, namely Z_BB is used to

either calculate the pressure given rho and T or the pressure. The code is as follows:

```

1  Function Z_BB(rho As Double, T As Double, param As Range, molw As Double, Optional opt As Double = 1)
2
3  'Calculates the compressibility factor given density and Temperature
4  '-----
5  'rho [=] Molar Density, kmol/m3
6  'T [=] Temperature, deg C
7  'Param [=] Excel Range that contains the numerical constants of the BB equation
8  'opt [=] optional selector to choose between Z or Pressure output (default is Z, opt = 1)
9  'Z_BB [=] Compressibility factor (calculated)
10 '*****
11
12 'Temp Changes
13 Dim Tk As Double
14 Tk = T + 273.15 'Temperature in K
15
16 'Density Changes from kmol/m3 to g/cm3
17 rho = rho * (molw / 1000)
18
19 'Retrieve the constants of the BB virial eq
20 '-----
21 Dim a As Double, Ao As Double, b As Double, Bo As Double, c As Double, d As Double
22 Ao = param.Cells(1, 1)
23 a = param.Cells(2, 1)
24 Bo = param.Cells(3, 1)
25 b = param.Cells(4, 1)
26 c = param.Cells(5, 1)
27 d = param.Cells(6, 1)
28
29 'Calculate beta, gamma delta
30 '-----
31 Dim beta As Double, gamma As Double, delta As Double, R As Double, R1 As Double, R2 As Double, r3 As
    ↪ Double
32 R = 2.93 ' cm3-atm/g-K
33 beta = Bo - (Ao / (R * Tk)) - (c / (Tk ^ 3))
34 gamma = ((a * Ao) / (R * Tk)) - (Bo * b) - ((Bo * c) / (Tk ^ 3))
35 delta = ((Bo * b * c) / (Tk ^ 3)) + ((Ao * d) / (R * Tk))
36
37 'Calculate the compressibility factor or Pressure
38 '-----
39 If opt = 1 Then
40 Z_BB = 1 + beta * rho + gamma * rho ^ 2 + delta * rho ^ 3
41 ElseIf opt = 2 Then
42 Z_BB = R * Tk * rho * (1 + beta * rho + gamma * rho ^ 2 + delta * rho ^ 3)

```

```

43 End If
44
45 End Function

```

Listing 1: VBA Function: Z_BB (Full Version)

Using this function (with opt set to 2 to return Pressure values), we were able to minimize our model pressure vs our experimental pressure at each point of the data we were given using the formula:

$$\Phi(d) = \min \sum_i (P_i^{\text{exp}} - P_i^{\text{model}})^2$$

By minimizing the sum squared difference (EXCEL SUMSQ) between the experimental pressure and the model pressure, we were able to find that the optimal value of d should be:

$$d = 1.410945$$

After plotting both model data when $d = 0$ vs experimental data and model data when $d = \text{optimal}$ vs experimental data, we were able to obtain both parity plots needed for this assignment, they are included below.

Parity Plot, $d = 0$

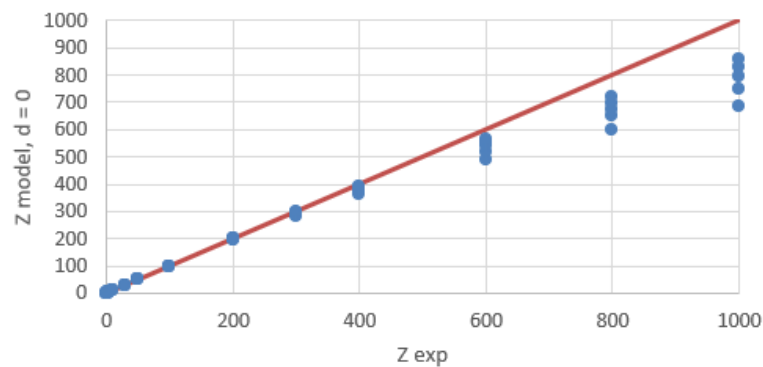


Figure 1: Z model vs Z experimental

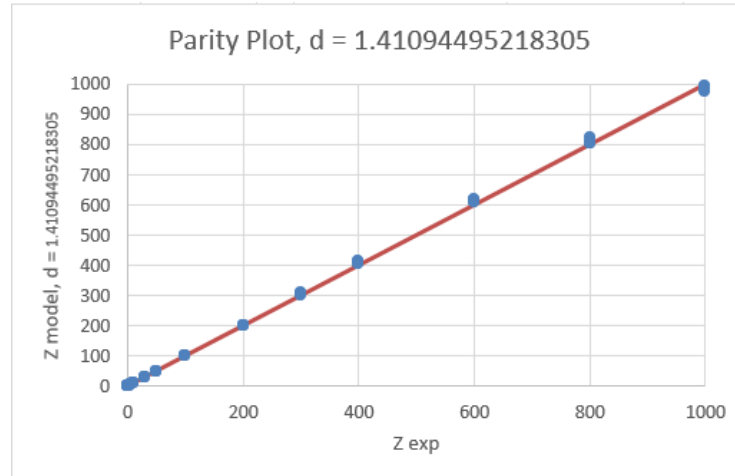


Figure 2: Z model opt vs Z experimental

We can very clearly see that when our d value is optimized, we get a much better parity plot and our data aligns to experimental values well. This concludes the optimization of d.

2. The second question asks us to calculate the thermal properties of Nitrogen using the Virial Equation. The first thermal property that we calculated was the thermal expansion coefficient, β_T . The general equation for the thermal expansion coefficient is:

$$\beta_T = -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial T} \right)_P = \frac{1}{\rho} \frac{\left(\frac{\partial P}{\partial T} \right)_\rho}{\left(\frac{\partial P}{\partial \rho} \right)_T}$$

where $\left(\frac{\partial P}{\partial T} \right)_\rho$ is the derivative of pressure with respect to temperature at constant density and $\left(\frac{\partial P}{\partial \rho} \right)_T$ is the derivative of pressure with respect to density at constant temperature. Using the equation solved for Pressure, P, found in Question 1, we can find $\left(\frac{\partial P}{\partial T} \right)_\rho$ and $\left(\frac{\partial P}{\partial \rho} \right)_T$:

$$\left(\frac{\partial P}{\partial T} \right)_\rho = \frac{\partial}{\partial T} [RT (\rho + \beta(T)\rho^2 + \gamma(T)\rho^3 + \delta(T)\rho^4)]$$

using the product rule, we get:

$$\left(\frac{\partial P}{\partial T} \right)_\rho = R [\rho + \beta(T)\rho^2 + \gamma(T)\rho^3 + \delta(T)\rho^4] + RT \left[\frac{\partial \beta}{\partial T} \rho^2 + \frac{\partial \gamma}{\partial T} \rho^3 + \frac{\partial \delta}{\partial T} \rho^4 \right]$$

where:

$$\begin{aligned} \frac{\partial \beta}{\partial T} &= \frac{3c}{T^4} + \frac{A_0}{RT^2} \\ \frac{\partial \gamma}{\partial T} &= \frac{3cB_o}{T^4} - \frac{aA_o}{RT^2} \\ \frac{\partial \delta}{\partial T} &= \frac{-3cbB_o}{T^4} - \frac{dA_o}{RT^2} \end{aligned}$$

and for $\left(\frac{\partial P}{\partial \rho} \right)_T$:

$$\left(\frac{\partial P}{\partial \rho} \right)_T = \frac{\partial}{\partial \rho} [RT (\rho + \beta(T)\rho^2 + \gamma(T)\rho^3 + \delta(T)\rho^4)]$$

$$\left(\frac{\partial P}{\partial \rho}\right)_T = RT [1 + 2\rho\beta(T) + 3\rho^2\gamma(T) + 4\rho^3\delta(T)]$$

We are now able to plug these equations into the thermal expansion coefficient equation to get:

$$\beta_T = \frac{1}{\rho} \frac{[(\rho + \beta(T)\rho^2 + \gamma(T)\rho^3 + \delta(T)\rho^4)] + T \left[\left(\frac{\partial \beta}{\partial T} \rho^2 + \frac{\partial \gamma}{\partial T} \rho^3 + \frac{\partial \delta}{\partial T} \rho^4 \right) \right]}{T [1 + 2\rho\beta(T) + 3\rho^2\gamma(T) + 4\rho^3\delta(T)]}$$

Solving this equation gets us:

$$\beta_T = 3.046 * 10^{-3} \frac{1}{K}$$

Now, we will solve for the heat capacity at constant volume, C_v . The general equation for C_v is:

$$C_v = C_v^{IG} + C_v^R$$

To solve for C_v , we first need to solve for C_v^R . The equation for C_v^R is:

$$C_v^R = \int_{\infty}^v \left(T \frac{\partial^2 P}{\partial T^2} \right)_v dv$$

This equation can be rewritten in terms of density:

$$C_v^R = \int_0^{\rho} \left(\frac{-T}{\rho^2} \frac{\partial^2 P}{\partial T^2} \right)_{\rho} d\rho$$

To find $\left(\frac{\partial^2 P}{\partial T^2}\right)_{\rho}$, we can take the derivative of $\left(\frac{\partial P}{\partial T}\right)_{\rho}$ that we found previously while solving for the thermal expansion coefficient:

$$\begin{aligned} \left(\frac{\partial P}{\partial T}\right)_{\rho} &= R [\rho + \beta(T)\rho^2 + \gamma(T)\rho^3 + \delta(T)\rho^4] + RT \left[\frac{\partial \beta}{\partial T} \rho^2 + \frac{\partial \gamma}{\partial T} \rho^3 + \frac{\partial \delta}{\partial T} \rho^4 \right] \\ \left(\frac{\partial^2 P}{\partial T^2}\right)_{\rho} &= \frac{\partial}{\partial T} R (\rho + \beta(T)\rho^2 + \gamma(T)\rho^3 + \delta(T)\rho^4) + \frac{\partial}{\partial T} RT \left(\frac{\partial \beta}{\partial T} \rho^2 + \frac{\partial \gamma}{\partial T} \rho^3 + \frac{\partial \delta}{\partial T} \rho^4 \right) \end{aligned}$$

Using the product rule, we get:

$$\begin{aligned} \left(\frac{\partial^2 P}{\partial T^2}\right)_{\rho} &= R \left[\frac{\partial \beta}{\partial T} \rho^2 + \frac{\partial \gamma}{\partial T} \rho^3 + \frac{\partial \delta}{\partial T} \rho^4 \right] + R \left[\frac{\partial \beta}{\partial T} \rho^2 + \frac{\partial \gamma}{\partial T} \rho^3 + \frac{\partial \delta}{\partial T} \rho^4 \right] + RT \left[\frac{\partial^2 \beta}{\partial T^2} \rho^2 + \frac{\partial^2 \gamma}{\partial T^2} \rho^3 + \frac{\partial^2 \delta}{\partial T^2} \rho^4 \right] \\ \left(\frac{\partial^2 P}{\partial T^2}\right)_{\rho} &= 2R \left[\frac{\partial \beta}{\partial T} \rho^2 + \frac{\partial \gamma}{\partial T} \rho^3 + \frac{\partial \delta}{\partial T} \rho^4 \right] + RT \left[\frac{\partial^2 \beta}{\partial T^2} \rho^2 + \frac{\partial^2 \gamma}{\partial T^2} \rho^3 + \frac{\partial^2 \delta}{\partial T^2} \rho^4 \right] \\ \left(\frac{\partial^2 P}{\partial T^2}\right)_{\rho} &= R \left[\rho^2 \left(2 \frac{\partial \beta}{\partial T} + T \frac{\partial^2 \beta}{\partial T^2} \right) + \rho^3 \left(2 \frac{\partial \gamma}{\partial T} + T \frac{\partial^2 \gamma}{\partial T^2} \right) + \rho^4 \left(2 \frac{\partial \delta}{\partial T} + T \frac{\partial^2 \delta}{\partial T^2} \right) \right] \end{aligned}$$

Where:

$$\frac{\partial^2 \beta}{\partial T^2} = \left(\frac{-12c}{T^5} - \frac{2A_0}{RT^3} \right)$$

$$\frac{\partial^2 \gamma}{\partial T^2} = \left(\frac{-12cB_o}{T^5} + \frac{2aA_o}{RT^3} \right)$$

$$\frac{\partial^2 \delta}{\partial T^2} = \left(\frac{12cbB_o}{T^5} + \frac{2dA_o}{RT^3} \right)$$

Now plugging in $\left(\frac{\partial^2 P}{\partial T^2}\right)_\rho$ into the equation of for C_v^R :

$$C_v^R = -RT \int_0^\rho \left[\left(2 \frac{\partial \beta}{\partial T} + T \frac{\partial^2 \beta}{\partial T^2} \right) + \rho \left(2 \frac{\partial \gamma}{\partial T} + T \frac{\partial^2 \gamma}{\partial T^2} \right) + \rho^2 \left(2 \frac{\partial \delta}{\partial T} + T \frac{\partial^2 \delta}{\partial T^2} \right) \right] d\rho$$

We can now use the CHE Calculator to find C_v^{IG} :

$$C_v^{IG} = 21.048 \frac{kJ}{kmol \cdot K}$$

Now solving for C_v , we get:

$$C_v = C^{IG} + C_v^R = 34.00 \frac{kJ}{kmol \cdot K}$$

The next thermal property we solved for was the heat capacity at constant pressure, C_P . The general equation for C_P is:

$$C_P = \frac{T}{\rho^2} \frac{\left[\left(\frac{\partial P}{\partial T} \right)_\rho \right]^2}{\left(\frac{\partial P}{\partial \rho} \right)_T} + C_V$$

Reusing the equations for $\left(\frac{\partial P}{\partial T}\right)_\rho$ and $\left(\frac{\partial P}{\partial \rho}\right)_T$ we derived earlier and our found value of the C_v , we get a value for C_P :

$$C_P = 53.03 \frac{kJ}{kmol \cdot K}$$

The next thermal property we calculated was the Joule-Thomson Coefficient, μ_{JT} . The general equation for the Joule-Thomson Coefficient is:

$$\mu_{JT} = \frac{T \cdot \beta_T - 1}{\rho \cdot C_P}$$

Using our previously calculated values for the thermal expansion coefficient, β_T , and the heat capacity at constant pressure, C_P , we find the value of the Joule-Thomson Coefficient to be:

$$\mu_{JT} = -1.00 * 10^{-7} \frac{K}{Pa}$$

The last thermal property that we calculated was the speed of sound, V_{sound} . The general equation for the speed of sound is:

$$V_{sound} = \sqrt{\frac{1}{MW} \frac{C_P}{C_V} \left(\frac{\partial P}{\partial \rho} \right)_T}$$

Reusing the previously derived equation for $\left(\frac{\partial P}{\partial \rho}\right)_T$ and the calculated values for the C_P and C_v along with the molecular weight of Nitrogen, (MW), we find the value of the speed of sound to be:

$$V_{\text{sound}} = 847.26 \frac{\text{m}}{\text{s}}$$

Using the above derivations, a VBA function was programmed that returns all of the secondary thermal properties given some key variables. We programmed it to return the Cp, the Cv, the Speed of Sound, the Joule-Thompson Coefficient, and the Thermal Expansion Coefficient. This function that we developed can also calculate enthalpy and entropy; one must simply change the opt value to 2 in the function inputs. The code is as follows:

```

1  Function ThermProps(rho As Double, T As Double, param As Range, molfrac As Range, molw As Double,
    ↪ Optional P As Double = 0, Optional opt As Integer = 1) As Variant
2
3  'Calculates the Secondary thermal properties given density and Temperature, etc...
4  '-----
5  'rho [=] Molar Density, kmol/m3
6  'T [=] Temperature, deg C
7  'Param [=] Excel Range that contains the numerical constants of the BB equation
8  'molfrac [=] Identity of the substance being evaluated
9  'molw [=] The molar weight of the substance being evaluated in kg/kmol
10 'P [=] optional, P in atm, needed for entropy calculation
11 'opt [=] optional, selector for secondary properties, enthalpy/entropy, or dpres_temp/dpres2_temp,
    ↪ default is 1
12 'ThermProps [=] Secondary Thermal Properties (Calculated)
13 '*****
14
15 'Define array that contains final results
16 Dim outputarray1(1 To 5, 1 To 1) As Variant ' 5 rows, 1 column
17 Dim outputarray2(1 To 2, 1 To 1) As Variant
18 'Change rho from kmol/m3 to g/cm3
19 rho = rho * (molw / 1000)
20
21 'Define and grab parameters for calculation
22 Dim a As Double, Ao As Double, b As Double, Bo As Double, c As Double, d As Double, R As Double
23 Ao = param.Cells(1, 1)
24 a = param.Cells(2, 1)
25 Bo = param.Cells(3, 1)
26 b = param.Cells(4, 1)
27 c = param.Cells(5, 1)
28 d = param.Cells(6, 1)
29
30 'Temperature Changes to Kelvin

```

```

31 Dim Tk As Double
32 Tk = T + 273.15 'Temperature in K
33
34 'Universal Gas Constant used in this code
35 R = 2.93 'cm3-atm/g-K
36
37 'Calculation of numbers used for thermal property calculation
38 Dim beta As Double, gamma As Double, delta As Double, R1 As Double, R2 As Double
39
40 beta = Bo - (Ao / (R * Tk)) - (c / (Tk ^ 3))
41 gamma = ((a * Ao) / (R * Tk)) - (Bo * b) - ((Bo * c) / (Tk ^ 3))
42 delta = ((Bo * b * c) / (Tk ^ 3)) + ((Ao * d) / (R * Tk))
43
44 Dim betad As Double, gammad As Double, deltad As Double
45
46 betad = (3 * c / Tk ^ 4) + (Ao / R / Tk ^ 2) 'First derivative of beta
47 gammad = (3 * Bo * c / Tk ^ 4) - (Ao * a / R / Tk ^ 2) 'First derivative of gamma
48 deltad = (-3 * c * Bo * b / Tk ^ 4) - (Ao * d / R / Tk ^ 2) 'First derivative of delta
49
50 Dim betad2 As Double, gammad2 As Double, deltad2 As Double, dpres_rho As Double, dpres_temp As Double,
    ↪ dpres2_temp As Double
51
52 betad2 = (-12 * c / Tk ^ 5) - (2 * Ao / R / Tk ^ 3) 'Second derivative of beta
53 gammad2 = (-12 * Bo * c / Tk ^ 5) + (2 * a * Ao / R / Tk ^ 3) 'Second derivative of gamma
54 deltad2 = (12 * c * Bo * b / Tk ^ 5) + (2 * Ao * d / R / Tk ^ 3) 'Second derivative of delta
55
56 'Calculation of derivative of pressure with respect to temperature at constant rho
57 dpres_rho = R * (rho + beta * (rho ^ 2) + _
58     gamma * (rho ^ 3) + delta * (rho ^ 4)) + R * Tk * (betad * (rho ^ 2) + gammad * (rho ^ 3) + deltad
    ↪ * (rho ^ 4))
59
60 'Calculation of derivative of pressure with respect to rho at constant temperature
61 dpres_temp = R * Tk * (1 + 2 * rho * beta + 3 * (rho ^ 2) * gamma + 4 * (rho ^ 3) * delta)
62 dpres2_temp = R * Tk * (2 * beta + 6 * rho * gamma + 12 * rho ^ 2 * delta)
63
64 'Cv ideal calculation setup
65 Dim CvFormula As String, componentNameString As String
66 componentNameString = molfrac.Address
67 CvFormula = "=CvIG(" & componentNameString & ", " & T & ")"
68
69 'Thermal Property Calculations
70 Dim ThermExpCoe As Double, C_constvol As Double, C_constpres As Double, JouleThompCoe As Double,
    ↪ SpeedSound As Double
71

```

```

72 ThermExpCoe = (1 / rho) * (((rho + beta * rho ^ 2 + gamma * rho ^ 3 + delta * rho ^ 4) _
73   + Tk * (betad * rho ^ 2 + gammad * rho ^ 3 + deltad * rho ^ 4)) / _
74   (Tk * (1 + 2 * rho * beta + 3 * rho ^ 2 * gamma + 4 * rho ^ 3 * delta)))
75
76 C_constvol = Application.Evaluate(CvFormula) + ((-R * Tk * (rho * (2 * betad + Tk * betad2) _
77   + 0.5 * rho ^ 2 * (2 * gammad + Tk * gammad2) + (1 / 3) * rho ^ 3 * (2 * deltad + Tk * deltad2)))
78   ↪ * 0.101325 * molw)
79
80 C_constpres = ((Tk / rho ^ 2) * (dpres_rho ^ 2 / dpres_temp) * 0.101325 * molw) + C_constvol
81
82 JouleThompCoe = ((ThermExpCoe * Tk - 1) / rho / C_constpres) / 101325
83
84 SpeedSound = Sqr((1 / molw) * (C_constpres / C_constvol) * (dpres_temp)) * Sqr(1013250) * Sqr(molw) /
85   ↪ 100
86
87 'Enthalpy and Entropy Calculations
88 Dim Enthalpy As Double, Entropy As Double, Z As Double, fugacitycoefficient As Double, H_reduced As Double,
89   ↪ S_reduced As Double
90 Dim HIGFormula As String
91 componentNameString = molfrac.Address
92 HIGFormula = "=HIG(" & componentNameString & ", " & T & ")"
93 Dim SIGFormula As String
94 componentNameString = molfrac.Address
95 SIGFormula = "=sig(" & componentNameString & ", " & T & ", " & P & ")"
96
97 Z = 1 + beta * rho + gamma * rho ^ 2 + delta * rho ^ 3
98 H_reduced = R * Tk * (Z - 1) - R * Tk ^ 2 * (betad * rho + 0.5 * gammad * rho ^ 2 + (1 / 3) * deltad *
99   ↪ rho ^ 3)
100
101 Enthalpy = Application.Evaluate(HIGFormula) + H_reduced * 0.101325 * molw
102 fugacitycoefficient = (Z - 1) - Log(Z) + (beta * rho + 0.5 * gamma * rho ^ 2 + (1 / 3) * delta * rho ^ 3)
103 S_reduced = R * (H_reduced / R / Tk - fugacitycoefficient)
104 Entropy = Application.Evaluate(SIGFormula) + S_reduced * 0.101325 * molw
105
106 If opt = 1 Then
107   'Final assignment of properties to output array
108   outputarray1(1, 1) = C_constpres 'kJ/kmol-K
109   outputarray1(2, 1) = C_constvol 'kJ/kmol-K
110   outputarray1(3, 1) = SpeedSound 'm/s
111   outputarray1(4, 1) = JouleThompCoe 'K/Pa
112   outputarray1(5, 1) = ThermExpCoe '1/K
113
114   'Final output
115   ThermProps = outputarray1
116 ElseIf opt = 2 Then

```

```

112 outputarray2(1, 1) = Enthalpy
113 outputarray2(2, 1) = Entropy
114
115 'Final output
116 ThermProps = outputarray2
117 End If
118 If opt = 3 Then
119 outputarray2(1, 1) = dpres_temp
120 outputarray2(2, 1) = dpres2_temp
121
122 'Final output
123 ThermProps = outputarray2
124 End If
125
126 End Function

```

Listing 2: VBA Function: ThermProps (Full Version)

This code calculates all of the values using the derivations above. The CvIG, HIG, and SIG are all calculated using the CHE Calculators' built-in functions. Using this function, we were able to calculate all of the properties at -60 °C.

Next, we were also tasked with calculating the secondary thermal values at -60 °C using the RKS EOS as well. To do this, we simply implemented the equations that were already derived in the book. Taking these equations, we wrote the following code:

```

1  Function RKS_props(molfrac As Range, T As Double, Tc As Double, P As Double, Pc As Double, w As Double
    ↪ , molw As Double, Optional opt As Integer = 1) As Variant
2  'Calculates the Secondary thermal properties given density and Temperature, etc...
3  '-----
4  'molfrac [=] Identity of the substance being evaluated
5  'T [=] Temperature, deg C
6  'Tc [=] Critical temperature, deg C
7  'P [=] Pressure, atm
8  'Pc [=] Critical Pressure, atm
9  'w [=] Acentric Factor
10 'molw [=] The molar weight of the substance being evaluated in kg/kmol
11 'opt [=] selector between secondary thermal properties, Enthalpy/Entropy, Z or rho, default is 1
12 'RKS_props [=] Secondary Thermal Properties (Calculated)
13 '*****
14
15 'Definition of output array and some key variables
16 If opt = 1 Then
17 Dim outputarray1(1 To 5, 1 To 1) As Variant ' 5 rows, 1 column
18 ElseIf opt = 2 Then

```

```

19 Dim outputarray2(1 To 2, 1 To 1) As Variant
20 End If
21
22 Dim a As Double, b As Double, lam1 As Double, lam2 As Double, sig As Double, kw As Double, _
23     Pr As Double, Tr As Double, Z As Double, Tk As Double, Acap As Double, Bcap As Double, R As Double
24
25 'Change of temperature from C to K
26 Tk = T + 273.15 'T in kelvin
27 Tc = Tc + 273.15 'Tc in Kelvin
28
29 'RKS constants defintion
30 lam1 = 0.42748
31 lam2 = 0.08664
32
33 'Universal Gas Constant
34 R = 2.93 'cm3-atm/g-K
35
36 'Calculation of reduced temp and pres
37 Tr = Tk / Tc
38 Pr = P / Pc
39
40 'Calculation of a and b for RKS
41 a = lam1 * (R * Tc) ^ 2 / Pc
42 b = lam2 * R * Tc / Pc
43
44 'Calculation of kw and sigma for RKS
45 kw = 0.48508 + 1.55171 * w - 0.15613 * w ^ 2
46 sig = (1 + kw * (1 - Sqr(Tr))) ^ 2
47
48 'Calculation of A and B for RKS
49 Acap = lam1 * sig * Pr / Tr ^ 2
50 Bcap = lam2 * Pr / Tr
51
52 'Code to solve for Cubic roots using function CubicSolver
53 Dim coeffA As Double, coeffB As Double, coeffC As Double
54 Dim roots As Variant, i As Integer
55
56 coeffA = 1
57 coeffB = -1
58 coeffC = Acap - Bcap - Bcap ^ 2
59 Dim coeffD As Double
60 coeffD = -Acap * Bcap
61
62 ' Solve cubic equation Z^3 + coeffB*Z^2 + coeffC*Z + coeffD = 0

```

```

63 roots = CubicSolver(coeffA, coeffB, coeffC, coeffD)
64
65 ' Pick the largest real root (vapor phase)
66 Z = Application.WorksheetFunction.Max(roots)
67
68 'rho calculation
69 Dim rho As Double
70 rho = P / R / Tk / Z
71
72 'Derivative Calculations
73 Dim D1 As Double, D2 As Double, C1 As Double, C2 As Double
74 D1 = -kw * Sqr(Tr / sig)
75 C1 = Z + (1 - D1) * (Acap / (Bcap + Z))
76 C2 = Z + (Bcap * Z / (Z - Bcap) ^ 2) - (Acap * Z / (Bcap + Z) ^ 2)
77 D2 = kw * (kw + 1) * Tr / 2 / sig / Sqr(Tr)
78
79 'Secondary Property Calculations
80 Dim CvFormula As String, componentNameString As String, C_constv As Double, C_constp As Double,
    ↪ beta_therm As Double, _
81 JT As Double, ss As Double
82
83 'Cv Ideal calculation preparation
84 componentNameString = molfrac.Address
85 CvFormula = "=CvIG(" & componentNameString & ", " & T & ")"
86
87 C_constv = Application.Evaluate(CvFormula) + (R * D2 * Acap / Bcap * Log(1 + Bcap / Z) * 0.101325 *
    ↪ molw)
88 C_constp = ((R * C1 ^ 2 / C2) * 0.101325 * molw) + C_constv
89 beta_therm = 1 / Tk * C1 / C2
90 JT = ((beta_therm * Tk - 1) / rho / C_constp) / 101325
91 ss = Sqr(R * Tk * C_constp / C_constv * C2) * Sqr(101.325)
92
93 'Enthalpy and Entropy Calculations
94 Dim Enthalpy As Double, Entropy As Double
95 Dim HIGFormula As String
96 componentNameString = molfrac.Address
97 HIGFormula = "=HIG(" & componentNameString & ", " & T & ")"
98 Dim SIGFormula As String
99 componentNameString = molfrac.Address
100 SIGFormula = "=sig(" & componentNameString & ", " & T & ", " & ATMtoBAR(P) & ")"
101
102 Enthalpy = Application.Evaluate(HIGFormula) + ((Z - 1) + (D1 - 1) * (Acap / Bcap) * Log(1 + Bcap / Z))
    ↪ * 0.101325 * molw
103 Entropy = Application.Evaluate(SIGFormula) + (Log(Z - Bcap) + D1 * (Acap / Bcap) * Log(1 + Bcap / Z))

```

```

↩ * 0.101325 * molw
104
105
106 If opt = 1 Then
107   'Assignment of secondary thermal props to output array
108   outputarray1(1, 1) = C_constp
109   outputarray1(2, 1) = C_constv
110   outputarray1(3, 1) = ss
111   outputarray1(4, 1) = JT
112   outputarray1(5, 1) = beta_therm
113   RKS_props = outputarray1
114 ElseIf opt = 2 Then
115   'Assignment of Enthalpy and Entropy to output array
116   outputarray2(1, 1) = Enthalpy
117   outputarray2(2, 1) = Entropy
118   RKS_props = outputarray2
119 End If
120 If opt = 3 Then
121   'assignment of Z to output
122   RKS_props = Z
123 End If
124 If opt = 4 Then
125   'assignment of rho to output
126   RKS_props = rho
127 End If
128
129 End Function

```

Listing 3: VBA Function: RKS_props (Full Version)

Now, you can see that in the code we use a CubicSolver function to solve for the roots of Z. This code was developed by Generative AI and is as follows:

```

1 Private Function CubeRoot(x As Double) As Double
2   ' --- Helper Function for CubicSolver: real cube root that handles negatives ---
3   If x >= 0# Then
4     CubeRoot = x ^ (1# / 3#)
5   Else
6     CubeRoot = -((-x) ^ (1# / 3#))
7   End If
8 End Function
9 Public Function CubicSolver(a As Double, b As Double, c As Double, d As Double) As Variant
10  ' --- CubicSolver: solves  $a*x^3 + b*x^2 + c*x + d = 0$  ---
11  'a [=] coef a
12  'b [=] coef b

```

```

13  'c [=] coef c
14  'd [=] coef d
15  'cubicsolver [=] returns roots of cubic equation
16
17  Dim roots(1 To 3) As Double
18  Dim EPS As Double: EPS = 0.000000000000001
19
20  ' Handle degenerate cases first
21  If Abs(a) < EPS Then
22      ' Reduce to quadratic: b*x^2 + c*x + d = 0
23      If Abs(b) < EPS Then
24          ' Linear: c*x + d = 0
25          If Abs(c) < EPS Then
26              roots(1) = 0#
27              roots(2) = 0#
28              roots(3) = 0#
29          Else
30              roots(1) = -d / c
31              roots(2) = 0#
32              roots(3) = 0#
33          End If
34          CubicSolver = roots
35          Exit Function
36      End If
37
38      Dim discQ As Double
39      discQ = c ^ 2 - 4# * b * d
40      If discQ >= 0# Then
41          roots(1) = (-c + Sqr(discQ)) / (2# * b)
42          roots(2) = (-c - Sqr(discQ)) / (2# * b)
43          roots(3) = 0#
44      Else
45          ' Return real part only (both complex)
46          roots(1) = -c / (2# * b)
47          roots(2) = 0#
48          roots(3) = 0#
49      End If
50      CubicSolver = roots
51      Exit Function
52  End If
53
54  ' --- Normalize coefficients for depressed cubic ---
55  Dim an As Double, bn As Double, cn As Double
56  an = b / a

```

```

57     bn = c / a
58     cn = d / a
59
60     ' Depressed cubic:  $x = y - an/3$  ?  $y^3 + p*y + q = 0$ 
61     Dim P As Double, q As Double
62     P = bn - (an ^ 2) / 3#
63     q = (2# * an ^ 3) / 27# - (an * bn) / 3# + cn
64
65     Dim discriminant As Double
66     discriminant = (q / 2#) ^ 2 + (P / 3#) ^ 3
67
68     If discriminant > EPS Then
69         ' One real root, two complex
70         Dim sqrtDisc As Double, u As Double, v As Double
71         sqrtDisc = Sqr(discriminant)
72         u = CubeRoot(-q / 2# + sqrtDisc)
73         v = CubeRoot(-q / 2# - sqrtDisc)
74         roots(1) = u + v - an / 3#
75         roots(2) = 0#
76         roots(3) = 0#
77     ElseIf Abs(discriminant) <= EPS Then
78         ' Multiple real roots
79         Dim u0 As Double
80         u0 = CubeRoot(-q / 2#)
81         roots(1) = 2# * u0 - an / 3#
82         roots(2) = -u0 - an / 3#
83         roots(3) = roots(2)
84     Else
85         ' Three distinct real roots (trig form)
86         Dim rho As Double, phi As Double, cosArg As Double
87         cosArg = (-q / 2#) / Sqr(-(P ^ 3) / 27#)
88         ' Clamp for safety
89         If cosArg > 1# Then cosArg = 1#
90         If cosArg < -1# Then cosArg = -1#
91         phi = Application.WorksheetFunction.Acos(cosArg)
92         rho = 2# * Sqr(-P / 3#)
93
94         roots(1) = rho * Cos(phi / 3#) - an / 3#
95         roots(2) = rho * Cos((phi + 2# * Application.WorksheetFunction.Pi()) / 3#) - an / 3#
96         roots(3) = rho * Cos((phi + 4# * Application.WorksheetFunction.Pi()) / 3#) - an / 3#
97     End If
98
99     CubicSolver = roots
100 End Function

```

Listing 4: VBA Function: CubicSolver (Full Version)

This CubicSolver takes any cubic equation and returns its roots. To find the correct root in our case, we simply take the max as it is the one for the vapor.

Using these codes, our final values for RKS and the Virial equation were compiled in SI units and were compared to the NIST values that were taken from the NIST Webbook, and the Peng-Robinson EOS. All of these values are compiled below.

Table 3: Comparison of Calculated Thermodynamic Properties

Property	Virial Eq Props	RKS	Peng-Rob Eq Props	NIST Webbook Props
C_p (J/mol·K)	53.02545521	43.21096022	41.87883627	41.604
C_v (J/mol·K)	34.00010837	27.59985394	26.1053147	25.109
SS (m/s)	731.8194538	781.0001824	745.4401864	847.26
JT (K/Pa)	-1.00056×10^{-7}	-1.5678×10^{-7}	-0.03795106	-4.8117×10^{-7}
β_{therm} (1/K)	0.003046084	0.002950519	0.002761605	n/a

We can see that our Virial EOS matches pretty well with all of our other three comparisons. In conclusion, our optimization of d was successful and we are able to calculate the secondary properties at different temperatures and pressures.

3. The first step for calculating the power to compress Nitrogen reversibly and adiabatically is to find the Entropy before and after the compression. To do this, we first need to find the Reduced Enthalpy, H^R , since the general equation for Entropy is:

$$S = S^{IG} + S^R$$

where S^{IG} is found using the CHE Calculator and the Reduced Entropy, S^R , is:

$$S^R = \frac{H^R}{T} - R \ln \varphi$$

To calculate the Reduced Enthalpy, we use the formula:

$$\frac{H^R}{RT} = (Z - 1) - \int_0^{\rho} \left(T \frac{\partial Z}{\partial T} \right)_{\rho} \frac{d\rho}{\rho}$$

To use this formula, we need to find $\left(\frac{\partial Z}{\partial T} \right)_{\rho}$. This is found by using our recast virial equation from Question 1:

$$Z = 1 + \beta(T)\rho + \gamma(T)\rho^2 + \delta(T)\rho^3$$

where:

$$\beta(T) = \left(-\frac{c}{T^3} + B_0 - \frac{A_0}{RT} \right)$$

$$\gamma(T) = \left(-\frac{B_0 c}{T^3} - B_0 b + \frac{A_0 a}{RT} \right)$$

$$\delta(T) = \left(\frac{cB_0b}{T^3} + \frac{A_0d}{RT} \right)$$

We then take the derivative of the virial equation with respect to Temperature, T, at constant Density, ρ :

$$\left(\frac{\partial Z}{\partial T} \right)_\rho = \frac{\partial \beta}{\partial T} \rho + \frac{\partial \gamma}{\partial T} \rho^2 + \frac{\partial \delta}{\partial T} \rho^3$$

where:

$$\begin{aligned} \frac{\partial \beta}{\partial T} &= \frac{3c}{T^4} + \frac{A_0}{RT^2} \\ \frac{\partial \gamma}{\partial T} &= \frac{3cB_0}{T^4} - \frac{aA_0}{RT^2} \\ \frac{\partial \delta}{\partial T} &= \frac{-3cbB_0}{T^4} - \frac{dA_0}{RT^2} \end{aligned}$$

We can now plug in the $\left(\frac{\partial Z}{\partial T} \right)_\rho$ formula in the equation for Reduced Enthalpy:

$$\frac{H^R}{RT} = (Z - 1) - T \int_0^\rho \frac{1}{\rho} \left[\frac{\partial \beta}{\partial T} \rho + \frac{\partial \gamma}{\partial T} \rho^2 + \frac{\partial \delta}{\partial T} \rho^3 \right] d\rho$$

After simplifying and solving for H^R , we get:

$$H^R = RT(Z - 1) - RT^2 \int_0^\rho \left[\frac{\partial \beta}{\partial T} + \frac{\partial \gamma}{\partial T} \rho + \frac{\partial \delta}{\partial T} \rho^2 \right] d\rho$$

Using this equation, we can also solve for the Enthalpy, since the general equation for Enthalpy is:

$$H = H^{IG} + H^R$$

Where H^{IG} is found using the CHE Calculator.

To find the Entropy, we also need to find the Fugacity Coefficient, $\ln \varphi$. This is given by the general equation:

$$\ln \varphi = (Z - 1) - \ln Z + \int_0^\rho (Z - 1) \frac{d\rho}{\rho}$$

where Z is our recast virial equation.

Using the Reduced Enthalpy equation allows us to find the Enthalpy of Nitrogen at $T_1 = -60^\circ C$ and $P_1 = 800 atm$ before compression. Using the Fugacity Coefficient Equation in conjunction with the Reduced Enthalpy allows us to find the Reduced Entropy, which in turn allows us to calculate the Entropy. Since the process is reversible, the Entropy before and after compression is the same ($S_1 = S_2$). Using this allows us to guess a temperature value for T_2 to use along with the given value of $P_2 = 1000 atm$. We also guessed a value for the density at the output of the compressor.

Finally, we simply implemented the code for enthalpy and entropy into the function ThermProps, adding simply another option for what it can calculate (Code is above in part 2). Using our function and the Newton Method to

balance our equations, we were able to calculate the final temperature that comes from compression as

$$T_f = -50.000\,521\,61\,^{\circ}\text{C}$$

At this temperature then, we were able to calculate the total work needed to compress as

$$W = 847.018\,181\,8\,\text{kJ kmol}^{-1}$$

We were also asked to compare this value to the one obtained from the RKS EOS. To do this, we again implemented the equations from our textbook into the RKS_props function, adding another option for what it can calculate (Full RKS_props code is written above in part 2). After implementing the code and calculating our final work at the same temperature, pressure and ρ as for the Virial Equation, we find that the total work is:

$$W = 361.391\,212\,8\,\text{kJ kmol}^{-1}$$

. Now this value does not align with the Virial Equation, but the Peng-Robinson EOS does yield a final work of:

$$W = 765.237\,652\,7\,\text{kJ kmol}^{-1}$$

which does align quite nicely with the Virial Equation. In conclusion then, we see that the virial equation we developed does align well with well known EOS like the Peng-Robinson EOS.

A table of our results is included below for the calculation of power needed for compression.

Table 4: Comparison of Thermodynamic Properties for Different Equations of State

Property	Virial		Peng–Robinson		RKS	
	S1	S2	S1	S2	S1	S2
Temperature ($^{\circ}\text{C}$)	-60.00	-50.00	-60.00	-49.84	-60.00	-47.72
Pressure (bar)	810.6	1013.25	810.6	1013.25	810.6	1013.25
ρ (g/cm^3)	0.6524	0.6872	25.8829	27.0471	0.6621	0.6889
ρ (kmol/m^3)	23.2899	24.5311	25.8829	27.0471	–	–
Equation balance	0.00	0.00	–	–	–	–
Z	1.9634	2.2257	1.7672	2.0177	1.9346	2.1985
Enthalpy (kJ/kmol)	-4168.96	-3321.95	-4435.12	-3669.88	-2488.75	-2127.35
Entropy ($\text{kJ}/\text{kmol}\cdot\text{K}$)	-75.7037	-75.7037	-74.8849	-74.8849	-68.7878	-68.7878
Entropy Balance	0.00	–	0.00	–	0.00	–
Work (kJ/kmol)	847.02	–	765.24	–	361.39	–

4. To verify if the Beattie-Bridgeman Equation of State is consistent at the critical point, we use the first and second derivatives of the pressure equation we first calculated in Question 1:

$$P = RT [\rho + \beta(T)\rho^2 + \gamma(T)\rho^3 + \delta(T)\rho^4]$$

We then find the first and second derivatives of Pressure with respect to density at constant temperature:

$$\left(\frac{\partial P}{\partial \rho}\right)_T = RT [1 + 2\rho\beta(T) + 3\rho^2\gamma(T) + 4\rho^3\delta(T)]$$

$$\left(\frac{\partial^2 P}{\partial \rho^2}\right)_T = RT [2\beta(T) + 6\rho\gamma(T) + 12\rho^2\delta(T)]$$

where:

$$\begin{aligned}\frac{\partial \beta}{\partial T} &= \frac{3c}{T^4} + \frac{A_0}{RT^2} \\ \frac{\partial \gamma}{\partial T} &= \frac{3cB_o}{T^4} - \frac{aA_o}{RT^2} \\ \frac{\partial \delta}{\partial T} &= \frac{-3cbB_o}{T^4} - \frac{dA_o}{RT^2}\end{aligned}$$

The reason for finding these derivatives is to find how the Pressure Function of the Virial Equation we found in Question 1 behaves at the critical temperature and pressure. If the equation is consistent at the critical point, the first and second derivatives of the function would be equal to zero. Using the value of $T_c = -146^\circ C$, and coding in the second derivative into our ThermProps function (we already had the first derivative), we found the first derivative to be equal to:

$$\left(\frac{\partial P}{\partial \rho}\right)_{T_c} = 658.49$$

and the second derivative to be:

$$\left(\frac{\partial^2 P}{\partial \rho^2}\right)_{T_c} = 6059.91$$

The code for both of these derivations is included above in part 2. To isolate the needed values, another option was added to the code that spits out only the values desired (ex: if opt = 3, then spit out the enthalpy and entropy).

Finally, since the values of the two derivatives are not equal to zero, the Beattie-Bridgeman Equation of State is not consistent at the critical point. This inconsistency arises from the data used. The data used is far from the critical point and over a large range of Pressures and Temperatures, meaning that the constants (A_o, a, B_o, b, c, d) are not optimized for critical conditions. To rectify this inconsistency, we would need data close to the critical point to find new constants to use for the equation of state. Doing this will allow the virial equation to be consistent at the critical point.